

# Harmonized Dual Policy Improvement for Modelic Reinforcement Learning

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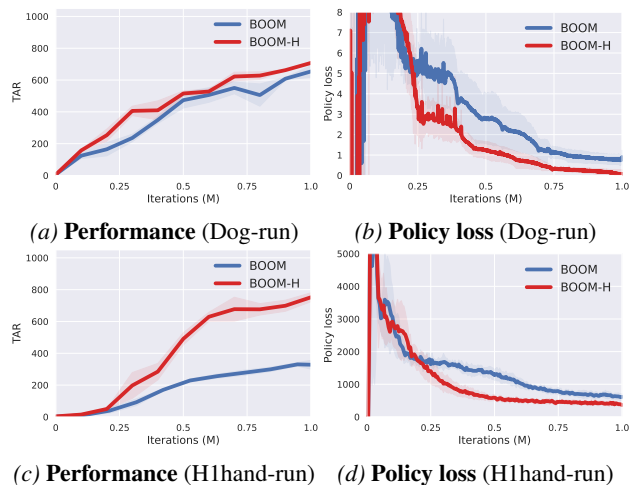
## Abstract

Policy-planner bootstrapping has emerged as a powerful paradigm in model-based reinforcement learning (MBRL). We formalize this process as a dual policy improvement mechanism synergizing: (i) *exploitative improvement* via off-policy  $Q$ -maximization, and (ii) *lookahead improvement* via planner alignment. While we theoretically prove that these improvements anchor to the same optimum, the practical training process inevitably encounters gradient inconsistency. Exacerbated by approximation inaccuracies and non-stationary data, this inconsistency induces destructive interference in policy updates, destabilizing the bootstrapping loop and leading to suboptimal convergence. To address this, we propose harmonized dual policy improvement (HDPI), a gradient-level framework that reconciles exploitative and lookahead improvements through a harmonized optimization scheme. This scheme effectively maximizes the worst-case inner product between the harmonized update and the original gradients, ensuring directional consistency and stabilizing policy evolution. Extensive empirical evaluations on 14 challenging tasks from the DeepMind Control Suite and the Humanoid-Bench demonstrate that HDPI significantly enhances training stability and asymptotic performance, outperforming a wide range of strong baselines.

## 1. Introduction

Model-based reinforcement learning (MBRL) has established itself as a cornerstone of sample-efficient control (Sutton, 1991; Hafner et al., 2019a; 2025), primarily through the use of world models that enable agents to plan within learned dynamics (Campbell et al., 2019; Moerland et al.,

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**Figure 1. Empirical evidence of reconciling the inconsistency between improvements.** Our proposed HDPI (red) effectively harmonizes the potential inconsistency between exploitative and lookahead improvements, ensuring stable policy updates. Conversely, the direct combination baseline (blue) suffers from destructive interference, compromising training stability and performance.

2023; Zhan et al., 2023; Hansen et al., 2023).

Among various architectures (Chua et al., 2018; Hafner et al., 2019b; Hamrick et al., 2020; Moerland et al., 2023), the policy-planner bootstrapping paradigm has recently set new records by elegantly combining the instantaneous response of a parametric policy with the multi-step lookahead of a model-based planner (Wang et al., 2025; Zhan et al., 2025a; Lin et al., 2025). This bidirectional synergy, where the planner guides the policy and the policy serves as a search prior, allows agents to master increasingly high-dimensional locomotion tasks (Zhan et al., 2025a). Despite the remarkable success achieved by the bootstrapping loop between the policy and the planner, its theoretical underpinnings remain somewhat elusive, resulting in a notable discrepancy between empirical success and its theoretical foundations.

To bridge this divide, we move beyond heuristic observations and formalize this interaction as a Dual Policy Improvement mechanism. Viewed through this lens, policy evolution is driven by two distinct forces: (i) *Exploitative Improvement*, which employs off-policy  $Q$ -maximization

for rapid, greedy policy refinement (Zhan et al., 2024b), and (ii) *Lookahead Improvement*, which harnesses the planner’s online multi-step search to ensure stable and far-sighted progress (Hansen et al., 2022). While we theoretically prove that these improvements anchor to the same optimum, practical training process inevitably encounters gradient inconsistency, often pointing in divergent or even opposing directions within the parameter space (Thrun & Schwartz, 2014; Liu et al., 2022; Zhan et al., 2025b). This inconsistency is not merely a transient artifact; it can be systematically exacerbated by function approximation inaccuracies, inherent model bias, and the non-stationary data distributions encountered during training (Melo et al., 2008; Wang et al., 2022; 2026). When these discordant gradients are naively combined, the resulting update signal is highly likely to be erratic (Yu et al., 2020; Sener & Koltun, 2018). This destabilizes the bootstrapping loop, leading to a “chasing-your-own-tail” problem where the policy and value function fail to co-evolve, ultimately resulting in suboptimal convergence or training collapse.

To address this, we propose Harmonized Dual Policy Improvement (HDPI), a gradient-level framework that moves beyond heuristic weighting. HDPI introduces a harmonized scheme to reconcile exploitative and lookahead updating forces. Rather than heuristic weighting, our mechanism seeks an update direction that maximizes the lower bound of projections onto both improvement directions. This strategy ensures that the “weaker” one is prioritized for enhancement, thereby securing for consistent policy improvement.

The key points of this paper are as follows:

- We formalize the policy-planner bootstrapping loop as a unified policy iteration framework termed dual policy improvement (DPI). Beyond conceptual unification, we provide rigorous theoretical guarantees for its convergence, establishing that both exploitative and lookahead mechanisms function as competent operators that drive the policy toward optimality.
- By establishing this dual-improvement foundation, we argue that simultaneous prioritization of both pathways is essential to unlock the synergy required for stable convergence. This perspective ensures that policy advancement is not merely the output of a single heuristic, but a coordinated pursuit of a shared optimum.
- We propose harmonized dual policy improvement (HDPI), a gradient-level framework designed to reconcile these dual mechanisms. By maximizing the worst-case projection (inner product) of the unified update onto the original improvement directions, HDPI analytically ensures directional consistency and mitigates destructive interference during policy evolution.

- Evaluations across 14 challenging tasks from the DeepMind Control Suite and Humanoid-Bench demonstrate that HDPI significantly outperforms a broad spectrum of competitive baselines, achieving superior performance and remarkably higher training stability.

## 2. Preliminaries

### 2.1. Off-Policy Reinforcement Learning

RL formulates sequential decision-making as a Markov Decision Process (MDP) defined by the tuple  $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ , where  $\mathcal{S}$  and  $\mathcal{A}$  are the state and action spaces,  $P(s'|s, a)$  is the transition dynamics,  $r(s, a)$  is the reward function, and  $\gamma \in [0, 1)$  is the discount factor (Li, 2023). The agent’s objective is to find an optimal policy  $\pi^*$  that maximizes the expected cumulative discounted return:  $J(\pi) = \mathbb{E}_\pi [\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)]$ . To quantify the cumulative return of a given policy  $\pi$ , one typically use state-value function  $V^\pi(s)$ , which is defined as the expected discounted return starting from state  $s$  and following policy  $\pi$  thereafter:

$$V^\pi(s) = \mathbb{E}_\pi \left[ \sum_{k=0}^{\infty} \gamma^k r(s_{t+k}, a_{t+k}) \middle| s_t = s \right]. \quad (1)$$

A central framework for optimizing this objective is policy iteration, which consists of two alternating phases: policy evaluation and policy improvement. During policy evaluation, the agent estimates the action-value function  $Q^\pi(s, a)$ , which represents the expected return starting from state  $s$ , taking action  $a$ , and thereafter following policy  $\pi$ :

$$Q^\pi(s, a) = \mathbb{E}_\pi \left[ \sum_{k=0}^{\infty} \gamma^k r(s_{t+k}, a_{t+k}) \right]. \quad (2)$$

In the policy improvement phase, the policy is updated to select actions that yield higher  $Q$ -values. In modern model-free off-policy RL, this is typically implemented by parameterizing the policy as  $\pi_\theta$  and updating  $\theta$  to maximize the estimated  $Q$ -function:

$$\mathcal{L}_{\max Q}(\theta) = \mathbb{E}_{s \sim \mathcal{D}} [-Q(s, \pi_\theta(s))], \quad (3)$$

where  $\mathcal{D}$  is a replay buffer of transitions. This iterative process ensures that the policy consistently moves toward higher-reward regions of the action space by exploiting the current value estimates.

### 2.2. Model-Based Reinforcement Learning

Unlike model-free approaches that primarily learn an actor and a critic, Model-based RL additionally learns a world model, typically comprising transition dynamics and reward functions, to provide predictive information that assists policy learning. A primary advantage of having a world

model is the ability to perform planning, also named lookahead search. Formally, given a state  $s_t$ , a dynamics model  $s_{t+1} = f(s_t, a_t)$ , a reward model  $r_t = R(s_t, a_t)$ , a terminal value function  $q_t = Q(s_t, a_t)$  and a planning horizon  $H$ , the planner, denoted as  $\pi_{s_t}^H$ , searches for an action sequence  $a_{t:t+H}$  that maximizes the predicted discounted return:

$$J_{s_t}^H(\pi) = \mathbb{E}_{\mathbf{a}_t \sim \pi} \left[ \sum_{i=0}^{H-1} \gamma^i \cdot r(s_{t+i}, \mathbf{a}_{t+i}) + V^{\pi_{old}}(s_{t+H}) \right], \quad (4)$$

Here  $\pi_{old}$  means the current parametric policy. Since off-policy RL algorithms typically learn the action-value function  $Q^{\pi_{old}}$ , therefore  $Q(s, a)^{\pi_{old}}$  is commonly employed to approximate  $V^{\pi_{old}}(s)$ . Note that the notation  $\pi$  in  $J_{s_t}^H(\pi)$  indicates that the first  $H$  actions follow the given policy  $\pi$ . Under the principle of optimal planning,  $\pi_{s_t}^H$  stands as the optimal policy for this lookahead objective, i.e.,

$$\pi_{s_t}^H = \arg \max_{\pi} J_{s_t}^H(\pi). \quad (5)$$

Recently, an increasing number of MBRL methods have sought to leverage this lookahead capability to achieve superior performance. Among these, the most effective paradigm to date involves constructing a forward Kullback-Leibler (KL) loss during training to align the parametric policy with the planner’s refined output, expressed as

$$\mathcal{L}_{\min \text{KL}}(\theta) = \mathbb{E}_{s \sim \mathcal{D}} [D_{\text{KL}}(\pi_{s_t}^H(\cdot|s) \parallel \pi_{\theta}(\cdot|s))]. \quad (6)$$

In this setup, the planner is warm-started by the current policy  $\pi_{\theta}$ , which in turn is updated to track the planner’s refined actions. This interaction forms a bootstrapping loop, a self-reinforcing cycle where the policy and planner iteratively guide each other toward higher performance. Through this mechanism, the policy gradually internalizes the lookahead reasoning capability of the planner, ultimately yielding robust and far-sighted behavior.

### 3. Method

#### 3.1. Dual Policy Improvement Framework

During the self-reinforcing bootstrapping loop in MBRL, the evolution of the parametric policy  $\pi_{\theta}$  is driven by two complementary update mechanisms, a process we formalize as Dual Policy Improvement (DPI). The first mechanism optimizes the policy via direct maximization of the action-value function (**Max Q**), while the second involves distilling the predictive foresight of a planner by minimizing their divergence (**Min KL**). In this subsection, we provide theoretical justifications to demonstrate that both mechanisms independently satisfy the criteria for policy improvement. By establishing their theoretical equivalence within the optimization landscape, we show that both serve as principled drivers for advancing agent performance.

**(1) Exploitative PIM: Max Q.** The first mechanism follows the standard off-policy RL paradigm. Given an action-value critic  $Q^{\pi_{old}}$ , the policy  $\pi_{old}$  is updated to maximize the expected return. This greedy or soft-greedy improvement is theoretically guaranteed to yield a superior policy, as formalized below:

**Theorem 3.1** (Exploitative Policy Improvement). *Let  $\pi_{old}$  and  $\pi_{new}$  be any pair of policies such that, for all  $s \in \mathcal{S}$ :*

$$Q^{\pi_{old}}(s, \pi_{new}(s)) \geq V^{\pi_{old}}(s). \quad (7)$$

*Then the policy  $\pi_{new}$  must be uniformly as good as, or better than,  $\pi_{old}$ . That is, for all  $s \in \mathcal{S}$ :*

$$V^{\pi_{new}}(s) \geq V^{\pi_{old}}(s). \quad (8)$$

*Proof.* The core logic of the proof involves inductively applying the Bellman consistency equation, in conjunction with the local improvement inequalities specified in (7). By recursively expanding the value function, we establish the long-term monotonic improvement. A detailed derivation is provided in Appendix A.1.  $\square$

**(2) Lookahead PIM: Min KL.** Beyond local gradient-based optimization, the second mechanism leverages the predictive capacity of a world model to perform multi-step lookahead. By optimizing trajectories over a finite horizon, a model-based planner derives an enhanced action distribution that serves as a high-fidelity reference for policy improvement. The parametric policy  $\pi_{\theta}$  then internalizes this superior strategy through a divergence-minimization objective. This alignment process allows the policy to transcend local value estimates by distilling the structural information provided by the planner, which we formally establish as a valid policy improvement step in the following theorem:

**Theorem 3.2** (Lookahead Policy Improvement). *Let  $\pi_{s_t}^H$  denote the online planner derived from the objective in (4). Analogous to (7), we can derive an improvement condition*

$$J_{s_t}^H(\pi_{s_t}^H) \leq \mathbb{E}_{\mathbf{a}_t \sim \pi_{s_t}^H} \left[ r(s_t, \mathbf{a}_t) + \gamma \cdot J_{s_{t+1}}^H(\pi_{s_{t+1}}^H) \right]. \quad (9)$$

*Let  $\pi_{new}$  be the new policy aligned with the planner, such that  $\pi_{new}(\cdot|s_t) = \pi_{s_t}^H(\cdot|s_t)$  for all  $s_t \in \mathcal{S}$ . Then, the new policy is uniformly as good as or better than the old policy:*

$$V^{\pi_{new}}(s) \geq V^{\pi_{old}}(s), \quad \forall s \in \mathcal{S}. \quad (10)$$

*Proof.* The core logic of the proof involves proving the local improvement inequalities specified in (9), and inductively applying the Bellman consistency equation, in conjunction with the (9). By recursively expanding the planning objective, we establish the long-term monotonic improvement. A detailed derivation is provided in Appendix A.2.  $\square$

**(3) Convergence to a Shared Optimum.** The dual objectives,  $\mathcal{L}_{\max Q}$  and  $\mathcal{L}_{\min KL}$ , represent synergistic pathways for policy refinement: (i)  $\mathcal{L}_{\max Q}$  drives improvement via the local gradient of the current value estimate, while (ii)  $\mathcal{L}_{\min KL}$  distills a high-fidelity search result guaranteed to be superior through  $H$ -step expansion. The following theorem establishes that these mechanisms are theoretically consistent, converging toward a unified optimal policy.

**Theorem 3.3** (Shared Optimum of Dual Policy Improvement). *Let  $\pi_0$  be an arbitrary initial policy. If the policy iteration using dual improvements is performed iteratively, the sequence of policies  $\{\pi_k\}_{k=0}^{\infty}$  converges to a unique optimal policy  $\pi_*$ . At the steady state, this optimal policy satisfies the following consistency relationship for all  $s_t \in \mathcal{S}$  and any arbitrary policy  $\pi$ :*

$$V^{\pi_*}(s) = J_s^H(\pi_*) = J_s^H(\pi_s^H) \geq J_s^H(\pi) \geq V^\pi(s). \quad (11)$$

*Proof.* The core logic of the proof involves applying the properties of optimal policy and optimal planning. A detailed derivation is provided in Appendix A.3.  $\square$

This theorem formally establishes that the exploitative gradient-based updates and the lookahead planning mechanism are perfectly aligned, converging to a consistent global optimum within the policy space.

### 3.2. Instability Risk in Dual Policy Improvement

While the dual policy improvement framework provides a principled foundation for policy-planner co-evolution, it introduces a critical source of instability: gradient inconsistency. This phenomenon occurs when the update signals derived from the two mechanisms point in divergent or even opposing directions within the parameter space.

Specifically, the exploitative gradient  $g_{\max Q} = \nabla_{\theta} \mathcal{L}_{\max Q}$  directs the policy toward actions that maximize the current  $Q$ -estimate, whereas the lookahead gradient  $g_{\min KL} = \nabla_{\theta} \mathcal{L}_{\min KL}$  pulls the policy toward the high-value strategies prescribed by the planner. In an idealized setting with an exact world model and a perfect value critic, these two signals would be perfectly aligned toward the global optimum. However, in practice, approximation errors in the learned world model, inherent bias in value estimation, and the non-stationarity of planner-generated targets often lead to a significant inconsistency between  $g_{\max Q}$  and  $g_{\min KL}$ .

A natural indicator of such inconsistency is the inner product between the two gradients. As a special case, a negative inner product  $\langle g_{\max Q}, g_{\min KL} \rangle < 0$  implies that the gradients form an obtuse angle, indicating a direct severe inconsistency where the two improvements compete for the same parameter updates. Such inconsistency has severe practical consequences: an update in the direction of one gradient may lead to a degradation of the other objective. This

”chasing-one’s-own-tail” dynamic destabilizes the policy-planner bootstrapping loop, as improvements in local value maximization may simultaneously impair the policy’s alignment with the planner’s multi-step foresight.

To quantitatively formalize the impact of this inconsistency, consider a parameter update  $\theta \leftarrow \theta - \eta h$  with learning rate  $\eta > 0$  and update direction  $h$ . The first-order change in each objective  $\mathcal{L}_i$  is given by:

$$\Delta \mathcal{L}_i = \mathcal{L}_i(\theta - \eta h) - \mathcal{L}_i(\theta) \approx -\eta \langle g_i, h \rangle. \quad (12)$$

For an objective to improve (i.e., for  $\mathcal{L}_i$  to decrease), the update direction  $h$  must maintain a positive inner product with its corresponding gradient ( $\langle h, g_i \rangle > 0$ ). Therefore, ideally, the inner product of the updated gradient  $h$  and the two gradients, i.e.,  $\langle h, g_{\max Q} \rangle$  and  $\langle h, g_{\min KL} \rangle$ , should be positive and as large as possible to better converge to the common optimal solution.

This gradient-level inconsistency represents a fundamental challenge that cannot be fully resolved through simple loss weighting alone. Even with a carefully tuned coefficient  $\lambda$ , local gradient mismatch can trigger oscillatory updates and degraded imitation performance, ultimately hindering the convergence toward the shared optimum.

### 3.3. Harmonized Dual Policy Gradient

To mitigate the aforementioned instability, we propose to explicitly harmonize the gradient signals at each optimization step. Rather than a naive weighted summation, we compute a Harmonized Dual Policy Gradient (HDPG) that minimizes the worst-case inconsistency while remaining anchored to the nominal update direction.

Formally, let  $\hat{g} = g_{\max Q} + \lambda g_{\min KL}$  be the nominal update direction, where  $\lambda$  is the alignment coefficient. To resolve gradient mismatch, we seek an update direction  $h \in \mathbb{R}^n$  that minimizes the maximum negative inner product with either gradient, subject to a trust-region constraint around the nominal direction:

$$\begin{aligned} \min_{h \in \mathbb{R}^n} \max_{i \in \{\max Q, \min KL\}} -\langle g_i, h \rangle \\ \text{s.t. } \|h - \hat{g}\| \leq \rho \|\hat{g}\|, \end{aligned} \quad (13)$$

where  $\rho \in (0, 1)$  is a harmonization coefficient that governs the trust-region radius. This minimax formulation has a clear geometric intuition: we search for an update  $h$  within a ball of radius  $\rho \|\hat{g}\|$  centered at  $\hat{g}$  that minimizes the most severe inconsistency with the original gradients. When  $\rho = 0$ ,  $h$  is constrained to  $\hat{g}$ , recovering the standard weighted update. As  $\rho$  increases, the framework gains the flexibility to pivot the update direction to alleviate inconsistency.

However, solving (13) directly is challenging due to its high-dimensional constrained optimization structure. Fortunately,

we can derive an equivalent one-dimensional problem that admits an efficient closed-form solution.

**Theorem 3.4** (Harmonized Dual Policy Gradient). *The solution to (13) is given by*

$$h = \hat{g} + \rho \frac{\|\hat{g}\|}{\|d(\alpha^*)\|} d(\alpha^*), \quad (14)$$

where  $d(\alpha) = \alpha g_{\max Q} + (2 - \alpha) g_{\min KL}$  for  $\alpha \in [0, 2]$ , and  $\alpha^*$  is obtained by solving the one-dimensional optimization:

$$\alpha^* = \arg \max_{\alpha \in [0, 2]} \langle d(\alpha), \hat{g} \rangle + \rho \|\hat{g}\| \|d(\alpha)\|. \quad (15)$$

The resulting  $h$  encapsulates information from both objectives, and is called *harmonized dual policy gradient*.

*Proof.* The core logic of the proof involves applying linear programming and Lagrangian duality. A detailed derivation is provided in Appendix B.  $\square$

Theorem 3.4 enables a highly efficient implementation of HDPG through drastic dimensionality reduction, as further elaborated in the following remark.

**Remark 3.5** (Efficient Implementation of HDPG). *The primal problem (13) is defined in the high-dimensional parameter space of the policy ( $\mathbb{R}^n$ ). Solving such a constrained minimax problem at every iteration would be computationally prohibitive. In contrast, the dualized problem (15) only requires optimizing a single scalar  $\alpha$ . This reduction drastically lowers the computational overhead, making harmonized gradient fusion practically feasible for complex tasks.*

### 3.4. Harmonized Dual Policy Improvement (HDPI)

Building upon the HDPG derivation, we propose Harmonized Dual Policy Improvement (HDPI), a framework that can be seamlessly integrated with existing planning-driven MBRL algorithms to achieve superior performance and enhanced training stability.

Specifically, this gradient-level harmonization operates as a *drop-in module*: it requires no structural modifications to the planner, the world model, or the underlying loss definitions. This modularity allows HDPI to be directly plugged into existing RL codebases with minimal effort. The overall procedure is summarized in Algorithm 1.

It is worth noting that the computational overhead introduced by HDPI is negligible. Solving for  $\alpha^*$  in (15) amounts to a simple one-dimensional line search over the interval  $[0, 2]$ , a cost that is significantly dwarfed by the training process, including back-propagation and online planning. By explicitly reconciling gradient inconsistency, HDPI effectively alleviates the instability stemming from neural network approximation errors and non-stationary data distributions, leading to more reliable and robust learning within the policy-planner bootstrap loop.

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#### Algorithm 1 Harmonized Dual Policy Improvement (HDPI)

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- 1: **Initialize:** Policy  $\pi_\theta$ , encoder  $h_\xi$ , dynamics  $f_\psi$ , reward  $R_\omega$ , value  $Q_\phi$ , planner  $\mathcal{P}$ , Replay buffer  $\mathcal{D}$
  - 2: **for** iteration = 1, 2, ... **do**
  - 3:   Collect experience and store in  $\mathcal{D}$
  - 4:   Update  $\{f_\psi, R_\omega, h_\xi, Q_\phi\}$  on mini-batch from  $\mathcal{D}$
  - 5:   Compute exploitative gradient:  $g_{\max Q} \leftarrow \nabla_\theta \mathcal{L}_{\max Q}$
  - 6:   Compute lookahead gradient:  $g_{\min KL} \leftarrow \nabla_\theta \mathcal{L}_{\min KL}$
  - 7:   Compute harmonized update direction  $h$  via (14)
  - 8:   Update policy:  $\theta \leftarrow \theta - \eta h$
  - 9: **end for**
- 

## 4. Experiments

### 4.1. Experimental Setup

**Benchmarks.** We adopted a challenging benchmark of 14 high-dimensional locomotion tasks drawn from the DeepMind Control Suite (DMC) (Tassa et al., 2018) and Humanoid Bench (H-Bench) (Sferrazza et al., 2024). The chosen 7 DMC tasks feature two most complex agents, humanoid (67/24 state/action dims) and dog (223/38), that demand sophisticated balance and coordination. The other 7 H-Bench tasks raise the difficulty further with long-horizon, goal-directed tasks on the Unitree H1 hand robot (151/61). Detailed descriptions are listed in Appendix C.1.

**Main Baselines.** We evaluate against 10 representative baselines: (1) **SAC** (Haarnoja et al., 2018): The standard maximum-entropy RL algorithm. (2) **DSAC-T** (Duan et al., 2024): A strong distributional RL algorithm that models Q-values as Gaussian distributions to address value overestimation. (3) **DreamerV3** (Hafner et al., 2025): The state-of-the-art imagination-driven algorithm that trains agents by rolling out a learned latent world model. (4) **TD-MPC2** (Hansen et al., 2023): A planning-driven algorithm that integrates task-oriented latent dynamics with an MPPI planner. (5) **BMPC** (Wang et al., 2025): An advanced variant of TD-MPC2 that learns the policy by solely imitating the planner’s actions with an online relabeling mechanism. (6) **BOOM** (Zhan et al., 2025a): A recent algorithm that establishes a closed policy-planner bootstrap loop, achieving a naive dual policy improvement with manually specified weights. We integrate our HDPI into this algorithm to establish our **BOOM-H** algorithm.

To provide a rigorous assessment against current high-performance paradigms, we additionally consider 4 recent diffusion RL algorithms as baselines: (7) **QVPO** (Ding et al., 2024), (8) **QSM** (Psenka et al., 2024), (9) **DACER** (Wang et al., 2024) and (10) **DPMD** (Ma et al., 2025), owing to their outstanding capabilities in diverse control tasks. A comprehensive introduction and additional empirical evidence can be found in Appendix C.2.

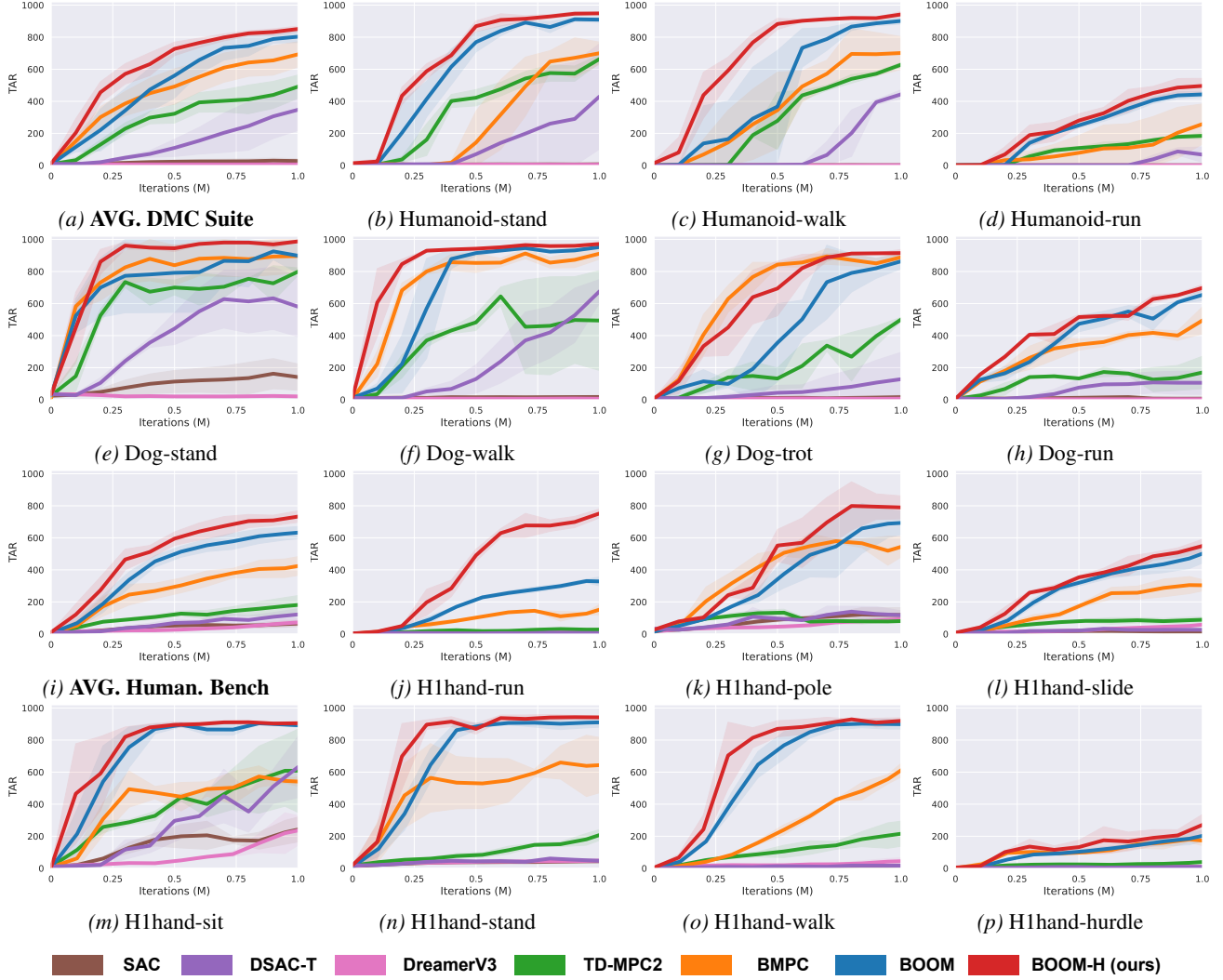


Figure 2. **Training curves on benchmarks.** The solid lines represent the mean, while the shaded regions indicate the confidence interval over three runs. The average performance curves for the two benchmarks appear at the left corner of the 1st and 3rd rows, respectively.

**Implementation details.** For all baselines, we utilize their officially released repositories and maintain their original hyperparameters without tuning to preserve their reported performance. Details are documented in Appendix C.3.

## 4.2. Experimental Results

Figure 2 showcases the comparative training evolution across all tasks and methods over a 1M-iteration horizon. Detailed numerical comparisons across 14 high-dimensional locomotion tasks are summarized in Table 1.

**Performance.** In DMC Suit, BOOM-H achieves a mean TAR of **852.6**, representing a **6.2%** improvement over the second-best method, BOOM (803.1), and a substantial **23.1%** gain over BMPC (692.8). These results validate that even in conventional benchmarks, HDPI can further

push the performance ceiling. The advantages of BOOM-H are even more pronounced on Humanoid-Bench, which involves the intricate dynamics of the H1 humanoid robot. BOOM-H attains a leading average TAR of **746.1**, marking a **16.8%** improvement over BOOM and a dramatic **70.4%** increase over BMPC. Notably, in the high-velocity *H1-Run* task, BOOM-H achieves a score of **752.1**, outperforming the second-best baseline by a significant margin of **127.7%**.

**Stability.** Beyond achieving record-breaking returns, a critical strength of the HDPI framework lies in its exceptional training stability. While baseline methods suffer from significant performance fluctuations and high standard deviations, BOOM-H maintains remarkably low variance across the majority of tasks. This stability indicates that HDPI effectively regularizes the policy evolution, ensuring robust and consistent convergence.

Table 1. **Total average return (TAR)** on 7 DMC Suite tasks and 7 Humanoid Benchmark (H-Bench) tasks. Mean  $\pm$  Std over 5 seeds. **Bold** = best, **Light Background** = second-best. Higher is better.

| DMC Suite     | H-Stand                           | H-Walk                             | H-Run                              | D-Stand                           | D-Walk                             | D-Trot                             | D-Run                              | Average                            |
|---------------|-----------------------------------|------------------------------------|------------------------------------|-----------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| SAC           | 7.4 $\pm$ 0.9                     | 2.4 $\pm$ 0.4                      | 1.3 $\pm$ 0.1                      | 162.2 $\pm$ 94.0                  | 17.1 $\pm$ 3.0                     | 16.0 $\pm$ 11.8                    | 16.5 $\pm$ 8.5                     | 29.7 $\pm$ 15.9                    |
| DSACT         | 429.6 $\pm$ 328.4                 | 443.4 $\pm$ 27.2                   | 86.8 $\pm$ 22.8                    | 632.8 $\pm$ 194.4                 | 675.8 $\pm$ 73.8                   | 127.9 $\pm$ 167.2                  | 107.8 $\pm$ 65.4                   | 347.2 $\pm$ 129.2                  |
| DreamerV3     | 5.5 $\pm$ 1.6                     | 1.6 $\pm$ 0.1                      | 0.8 $\pm$ 0.1                      | 35.4 $\pm$ 10.8                   | 7.6 $\pm$ 1.5                      | 7.9 $\pm$ 0.7                      | 4.3 $\pm$ 3.2                      | 8.4 $\pm$ 2.8                      |
| TD-MPC2       | 663.7 $\pm$ 26.5                  | 628.2 $\pm$ 13.3                   | 184.6 $\pm$ 19.4                   | 798.9 $\pm$ 23.3                  | 644.8 $\pm$ 57.8                   | 500.0 $\pm$ 13.6                   | 172.9 $\pm$ 29.5                   | 491.3 $\pm$ 72.6                   |
| BMPC          | 700.0 $\pm$ 69.6                  | 701.3 $\pm$ 102.6                  | 257.7 $\pm$ 125.6                  | 896.3 $\pm$ 105.7                 | 912.6 $\pm$ 32.4                   | 894.1 $\pm$ 49.0                   | 494.5 $\pm$ 84.2                   | 692.8 $\pm$ 78.9                   |
| BOOM          | 912.0 $\pm$ 13.0                  | 901.4 $\pm$ 19.9                   | 443.0 $\pm$ 18.2                   | 925.6 $\pm$ 51.4                  | 952.1 $\pm$ 4.4                    | 863.2 $\pm$ 46.6                   | 654.0 $\pm$ 36.1                   | 803.1 $\pm$ 35.9                   |
| <b>BOOM-H</b> | <b>948.6 <math>\pm</math> 2.2</b> | <b>942.0 <math>\pm</math> 18.5</b> | <b>496.1 <math>\pm</math> 47.0</b> | <b>987.6 <math>\pm</math> 3.8</b> | <b>971.6 <math>\pm</math> 13.4</b> | <b>915.0 <math>\pm</math> 22.2</b> | <b>707.5 <math>\pm</math> 10.7</b> | <b>852.6 <math>\pm</math> 16.8</b> |

| H-Bench       | H1-Stand                           | H1-Walk                           | H1-Run                             | H1-Sit                            | H1-Slide                           | H1-Pole                            | H1-Hurdle                          | Average                            |
|---------------|------------------------------------|-----------------------------------|------------------------------------|-----------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| SAC           | 44.1 $\pm$ 6.1                     | 16.1 $\pm$ 1.7                    | 9.3 $\pm$ 1.5                      | 259.6 $\pm$ 79.9                  | 19.0 $\pm$ 5.9                     | 121.6 $\pm$ 43.4                   | 9.1 $\pm$ 2.6                      | 67.3 $\pm$ 19.6                    |
| DSACT         | 61.0 $\pm$ 9.5                     | 17.0 $\pm$ 4.6                    | 8.7 $\pm$ 4.9                      | 631.4 $\pm$ 190.2                 | 32.6 $\pm$ 4.6                     | 139.2 $\pm$ 33.5                   | 8.9 $\pm$ 2.9                      | 121.9 $\pm$ 35.6                   |
| DreamerV3     | 50.8 $\pm$ 5.3                     | 46.6 $\pm$ 22.6                   | 14.0 $\pm$ 2.3                     | 252.1 $\pm$ 57.5                  | 66.0 $\pm$ 9.8                     | 98.3 $\pm$ 22.8                    | 11.0 $\pm$ 2.2                     | 77.0 $\pm$ 17.5                    |
| TD-MPC2       | 233.9 $\pm$ 30.3                   | 226.4 $\pm$ 77.3                  | 31.2 $\pm$ 3.0                     | 608.7 $\pm$ 221.7                 | 91.9 $\pm$ 14.7                    | 133.1 $\pm$ 21.3                   | 43.8 $\pm$ 11.6                    | 187.9 $\pm$ 63.9                   |
| BMPC          | 660.0 $\pm$ 103.1                  | 665.4 $\pm$ 55.7                  | 176.8 $\pm$ 54.6                   | 572.6 $\pm$ 66.2                  | 305.2 $\pm$ 53.1                   | 579.3 $\pm$ 11.5                   | 180.4 $\pm$ 18.5                   | 437.9 $\pm$ 58.7                   |
| BOOM          | 912.3 $\pm$ 28.1                   | 902.1 $\pm$ 25.2                  | 330.3 $\pm$ 11.3                   | 904.4 $\pm$ 4.4                   | 531.2 $\pm$ 55.4                   | 697.6 $\pm$ 56.9                   | 220.0 $\pm$ 29.0                   | 639.0 $\pm$ 36.5                   |
| <b>BOOM-H</b> | <b>942.8 <math>\pm</math> 11.9</b> | <b>929.4 <math>\pm</math> 1.6</b> | <b>752.1 <math>\pm</math> 27.9</b> | <b>911.6 <math>\pm</math> 7.3</b> | <b>610.5 <math>\pm</math> 33.2</b> | <b>823.2 <math>\pm</math> 39.0</b> | <b>270.8 <math>\pm</math> 65.0</b> | <b>746.1 <math>\pm</math> 30.0</b> |

**Task Abbreviation:** (1) **H-** prefix refers to the DMC *Humanoid* domain (e.g., H-Stand is Humanoid-Stand). (2) **D-** prefix refers to the DMC *Dog* domain (e.g., D-Trot is Dog-Trot). (3) **H1-** prefix denotes the *H1hand* in humanoid benchmark.

**Comparative Summary.** In contrast to the limited scalability of SAC and DSAC-T, the relative sample inefficiency of DreamerV3, and the characteristic instability of BMPC and TD-MPC2, BOOM-H exemplifies consistent superiority across the evaluation suite. By achieving the highest final returns coupled with the lowest variance across all 14 tasks, our framework substantiates that HDPI provides a robust, sample-efficient, and scalable solution for high-dimensional continuous control.

### 4.3. Supplementary Evidence and Analysis

We present empirical evidence to substantiate our claims regarding enhanced policy optimization and the practical efficiency of our implementation.

**Evidence of Policy Optimization Benefits.** To illustrate how our method facilitates the learning process, we analyze the training dynamics of BOOM and BOOM-H on two demanding tasks: *Dog-run* and *H1hand-run*. As shown in Figure 1, our framework produces a more stable and effective policy loss profile. Specifically, Figures 1b and 1d demonstrate that the policy loss in BOOM-H converges more rapidly and attains a lower steady-state value compared to the baseline. This optimized training objective directly translates into superior sample efficiency and higher asymptotic performance (Figures 1a and 1c). These results suggest that the proposed alignment between the planner and the policy effectively regularizes the optimization landscape, mitigating training instabilities.

**Computational Efficiency.** To evaluate the computational overhead of our approach, we measure the wall-clock time relative to the baseline. For a 10k-iteration training session, the execution time increases only marginally from 312 to 339 seconds. This minimal overhead (approximately 8%) confirms that our harmonization scheme is computationally lightweight. In practice, the system’s primary bottlenecks remain the neural network backpropagation and the trajectory sampling within the planner, rather than our auxiliary harmonization process.

### 4.4. Ablation Study

We select the *Dog-run* task from the DMC Suite, which features the highest dimensionality, as our primary task for the ablation experiments.

**Sensitivity to harmonization coefficient  $\rho$ .** We investigate the sensitivity of our framework to the harmonization coefficient  $\rho$ , which governs the extent of gradient harmonization. As illustrated in Figure 3, both the agent’s performance and policy loss trajectories exhibit remarkable stability across a wide range of  $\rho$  values. This empirical robustness suggests that the proposed harmonization technique benefits from a well-behaved optimization landscape and alleviates the need for exhaustive tuning.

**Impact of Alignment Coefficient  $\lambda$ .** The alignment coefficient  $\lambda$  determines the nominal balancing between  $g_{\max Q}$  and  $g_{\min KL}$ . We evaluate its impact by varying  $\lambda$  across

three orders of magnitude ( $0.1\times$ ,  $1\times$ , and  $10\times$  the default scale). The results in Figure 4 reveal that our harmonized approach consistently outperforms the original BOOM baseline across all tested scales in terms of asymptotic return. Such inherent robustness is particularly advantageous for deploying the algorithm across diverse environments with heterogeneous task configurations, demonstrating the superior generalizability of HDPI.

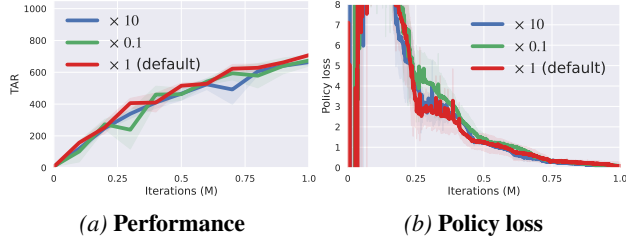


Figure 3. Ablation study on the harmonization coefficient  $\rho$ , demonstrating insensitivity to this introduced hyperparameter.

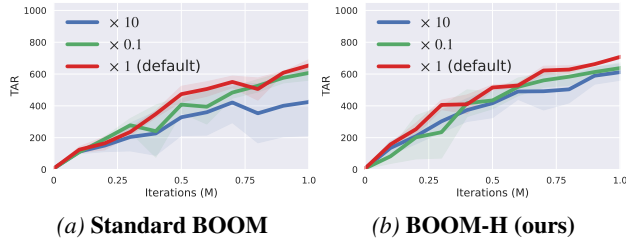


Figure 4. Ablation study on the alignment coefficient  $\lambda$ , highlighting the consistent gains of HDPI under varying settings.

## 5. Related Work

According to the mechanism of policy improvement, We categorize the existing literature into two dominant paradigms.

**Exploitative policy improvement.** This paradigm focuses on deriving policies from the jointly learning action-value functions (Zhan et al., 2026). The primary challenge lies in the accuracy of  $Q$ -estimation. While the vanilla SAC (Haarnoja et al., 2018) provides a robust maximum-entropy framework, it often suffers from overestimation bias. To address this, DSAC-T (Duan et al., 2024) models the  $Q$ -value as a Gaussian distribution to capture aleatoric uncertainty and stabilizes gradients through variance-based adjustments. Other works seek higher expressiveness or alternative optimization targets: QVPO (Ding et al., 2024) and DACER (Wang et al., 2024) leverage diffusion models as policy to capture high  $Q$ -values regions in a multi-modal manner, while QSM (Psenka et al., 2024) employs score-matching to align policy gradients directly with  $Q$ -gradients. Despite these advancements, these methods are exploitative solely relying on  $Q$ -value without explicit reasoning about internal environment transitions.

**Lookahead policy improvement.** Model-based RL often intensifies the dependence on world model fidelity, making predictive accuracy a critical determinant of performance (Zhan et al., 2024a). By learning a latent dynamics model, imagination-driven methods (Kaiser et al., 2019; Micheli et al., 2022; Mu et al., 2021; Hafner et al., 2019a; 2020), such as DreamerV3 (Hafner et al., 2025), learns policies through virtual rollouts, significantly enhancing sample efficiency. Alternatively, planning-driven methods (Curi et al., 2020; Sikchi et al., 2022; Chua et al., 2018), like TD-MPC2 (Hansen et al., 2023), perform online look-ahead search (e.g., MPPI) to find high-quality behaviors. By explicitly simulating future trajectories, these methods alleviate the reliance on global  $Q$ -function precision. However, they remain susceptible to the fidelity of the world model, as the policy tends to exploit predictive inaccuracies during optimization.

## Comparison with multi-task or multi-objective RL.

Our work structurally differs from multi-task (Vithayathil Varghese & Mahmoud, 2020) or multi-objective reinforcement learning (Mossalam et al., 2016) in terms of the optimization objective and the role of balancing techniques. In their settings, methods such as loss magnitude balancing (Chen et al., 2018; Liu et al., 2019) and gradient clipping (Yu et al., 2020; Sener & Koltun, 2018) are typically employed to navigate competing objectives or avoid dominant gradients, seeking a Pareto-optimal trade-off among potentially conflicting goals (Lin et al., 2019). While our framework also feels like stabilization techniques, their underlying purpose is fundamentally different: we want to ensure a *harmonized* joint optimization toward a single, shared optimal solution. Since the dual PIM mechanisms are mathematically grounded to converge to the same global optimum, our techniques serve to synchronize their improvement rates rather than to mediate between diverging tasks. Our harmonization ensures that the interaction between the policy and the planner consistently facilitates mutual refinement instead of addressing the gradient conflict common in multi-objective landscapes.

## 6. Conclusion

This paper theoretically formalizes policy-planner bootstrapping in MBRL as a dual policy improvement (DPI) framework and prove that their share a common optimum. To address the inevitable destructive gradient inconsistency during practical training, we propose Harmonized Dual Policy Improvement (HDPI), which reconciles these inconsistencies through a harmonization scheme. Extensive evaluations across 14 high-dimensional tasks demonstrate that HDPI significantly stabilizes the bootstrapping process and achieves state-of-the-art performance. As a modular, drop-in solution, HDPI offers a robust foundation for subsequent research on

policy-planner bootstrapping in MBRL.

## Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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## A. Theoretical Analysis of Dual Policy Improvement

In this appendix, we provide the detailed proofs for the two types of policy improvement discussed in Section 3.1. We begin by defining the necessary notation and lemmas.

### A.1. Proof of Exploitative PIM via Max Q

**Theorem 3.1:** *Let  $\pi_{old}$  and  $\pi_{new}$  be any pair of policies such that, for all  $s \in \mathcal{S}$ :*

$$Q^{\pi_{old}}(s, \pi_{new}(s)) \geq V^{\pi_{old}}(s). \quad (16)$$

*Then the policy  $\pi_{new}$  must be uniformly as good as, or better than,  $\pi_{old}$ . That is, for all  $s \in \mathcal{S}$ :*

$$V^{\pi_{new}}(s) \geq V^{\pi_{old}}(s). \quad (17)$$

*Proof.* Consider the initial condition  $V^{\pi_{old}}(s) \leq Q^{\pi_{old}}(s, \pi_{new}(s))$ . By the definition of the action-value function  $Q^{\pi_{old}}$ , we can expand the right-hand side as the expected sum of the immediate reward and the discounted value of the successor state:

$$\begin{aligned} V^{\pi_{old}}(s) &\leq Q^{\pi_{old}}(s, \pi_{new}(s)) \\ &= \mathbb{E}[r_{t+1} + \gamma V^{\pi_{old}}(s_{t+1}) \mid s_t = s, a_t = \pi_{new}(s)] \\ &= \mathbb{E}_{\pi_{new}}[r_{t+1} + \gamma V^{\pi_{old}}(s_{t+1}) \mid s_t = s] \end{aligned} \quad (18)$$

where  $\mathbb{E}_{\pi_{new}}[\cdot]$  denotes the expectation following policy  $\pi_{new}$ . We now iteratively apply the assumption  $V^{\pi_{old}}(s) \leq Q^{\pi_{old}}(s, \pi_{new}(s))$  to the state  $s_{t+1}$  on the right-hand side of Equation 18:

$$\begin{aligned} V^{\pi_{old}}(s) &\leq \mathbb{E}_{\pi_{new}}[r_{t+1} + \gamma Q^{\pi_{old}}(s_{t+1}, \pi_{new}(s_{t+1})) \mid s_t = s] \\ &= \mathbb{E}_{\pi_{new}}[r_{t+1} + \gamma \mathbb{E}_{\pi_{new}}[r_{t+2} + \gamma V^{\pi_{old}}(s_{t+2}) \mid s_{t+1}]] \\ &= \mathbb{E}_{\pi_{new}}[r_{t+1} + \gamma r_{t+2} + \gamma^2 V^{\pi_{old}}(s_{t+2}) \mid s_t = s] \end{aligned}$$

By continuing this recursive substitution for  $n$  steps, we obtain:

$$V^{\pi_{old}}(s) \leq \mathbb{E}_{\pi_{new}} \left[ \sum_{k=0}^{n-1} \gamma^k r_{t+k+1} + \gamma^n V^{\pi_{old}}(s_{t+n}) \mid s_t = s \right] \quad (19)$$

As  $n \rightarrow \infty$ , the remainder term  $\lim_{n \rightarrow \infty} \mathbb{E}_{\pi_{new}}[\gamma^n V^{\pi_{old}}(s_{t+n})] = 0$ . Thus, we are left with:

$$V^{\pi_{old}}(s) \leq \mathbb{E}_{\pi_{new}} \left[ \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = s \right] = V^{\pi_{new}}(s) \quad (20)$$

This establishes that  $V^{\pi_{new}}(s) \geq V^{\pi_{old}}(s)$  for all  $s \in \mathcal{S}$ .  $\square$

### A.2. Proof of Lookahead PIM via Min KL

**Theorem 3.2:** *Let  $\pi_{s_t}^H$  denote the online planner derived from the objective in Equation (4). Let  $\pi_{new}$  be the new policy aligned with the planner, such that  $\pi_{new}(\cdot \mid s_t) = \pi_{s_t}^H(\cdot \mid s_t)$  for all  $s_t \in \mathcal{S}$ . Then, the new policy is uniformly as good as or better than the old policy:*

$$V^{\pi_{new}}(s) \geq V^{\pi_{old}}(s), \quad \forall s \in \mathcal{S}. \quad (21)$$

*Proof.* Before the proof, we need to show that

$$V^{\pi_{old}}(s_t) \leq J_{s_t}^H(\pi_{s_t}^H), \quad (22)$$

because  $\pi_{s_t}^H$  is the optimal solution and  $V^{\pi_{old}}(s_t) = J_{s_t}^H(\pi_{old}) \leq J_{s_t}^H(\pi_{s_t}^H)$ .

Next, we will prove that

$$J_{s_t}^H(\pi_{s_t}^H) \leq \mathbb{E}_{\mathbf{a}_t \sim \pi_{s_t}^H} \left[ r(s_t, \mathbf{a}_t) + \gamma \cdot J_{s_{t+1}}^H(\pi_{s_{t+1}}^H) \right], \quad (23)$$

which follows

$$\begin{aligned}
 J_{s_t}^H(\pi_{s_t}^H) &= \mathbb{E}_{\mathbf{a}_{t:t+H-1} \sim \pi_{s_t}^H} [r(s_t, \mathbf{a}_t) + \dots + \gamma^{H-1} \cdot r(s_{t+H-1}, \mathbf{a}_{t+H-1}) + \gamma^H \cdot V^{\pi_{old}}(s_{t+H})] \\
 &= \mathbb{E}_{\mathbf{a}_{t:t+H-1} \sim \pi_{s_t}^H, \mathbf{a}_{t+H} \sim \pi_{old}} \left[ r(s_t, \mathbf{a}_t) + \dots + \gamma^{H-1} \cdot r(s_{t+H-1}, \mathbf{a}_{t+H-1}) \right. \\
 &\quad \left. + \gamma^H \cdot r(s_{t+H}, \mathbf{a}_{t+H}) + \gamma^{H+1} \cdot V^{\pi_{old}}(s_{t+H+1}) \right] \\
 &= \mathbb{E}_{\mathbf{a}_t \sim \pi_{s_t}^H, \mathbf{a}_{t+1:t+H-1} \sim \pi_{s_t}^H, \mathbf{a}_{t+H} \sim \pi_{old}} \left[ r(s_t, \mathbf{a}_t) + \gamma \cdot [r(s_{t+1}, \mathbf{a}_{t+1}) + \dots + \gamma^{H-2} \cdot r(s_{t+H-1}, \mathbf{a}_{t+H-1}) \right. \\
 &\quad \left. + \gamma^{H-1} \cdot r(s_{t+H}, \mathbf{a}_{t+H}) + \gamma^H \cdot V^{\pi_{old}}(s_{t+H+1})] \right] \\
 &\leq \mathbb{E}_{\mathbf{a}_t \sim \pi_{s_t}^H, \mathbf{a}_{t+1:t+H} \sim \pi_{s_{t+1}}^H} \left[ r(s_t, \mathbf{a}_t) + \gamma \cdot [r(s_{t+1}, \mathbf{a}_{t+1}) + \dots + \gamma^{H-1} \cdot r(s_{t+H-1}, \mathbf{a}_{t+H-1}) \right. \\
 &\quad \left. + \gamma^H \cdot V^{\pi_{old}}(s_{t+H+1})] \right] \\
 &= \mathbb{E}_{\mathbf{a}_t \sim \pi_{s_t}^H} \left[ r(s_t, \mathbf{a}_t) + \gamma \cdot J_{s_{t+1}}^H(\pi_{s_{t+1}}^H) \right].
 \end{aligned}$$

We finish the proof by applying Equation (22) and iteratively applying Equation (23):

$$\begin{aligned}
 V^{\pi_{old}}(s_t) &\leq J_{s_t}^H(\pi_{s_t}^H) \\
 &\leq \mathbb{E}_{\mathbf{a}_t \sim \pi_{s_t}^H} \left[ r(s_t, \mathbf{a}_t) + \gamma \cdot J_{s_{t+1}}^H(\pi_{s_{t+1}}^H) \right] \\
 &\leq \mathbb{E}_{\mathbf{a}_t \sim \pi_{s_t}^H, \mathbf{a}_{t+1} \sim \pi_{s_{t+1}}^H} \left[ r(s_t, \mathbf{a}_t) + \gamma \cdot r(s_{t+1}, \mathbf{a}_{t+1}) + \gamma^2 \cdot J_{s_{t+2}}^H(\pi_{s_{t+2}}^H) \right] \\
 &\leq \mathbb{E}_{\mathbf{a}_t \sim \pi_{s_t}^H, \mathbf{a}_{t+1} \sim \pi_{s_{t+1}}^H, \mathbf{a}_{t+2} \sim \pi_{s_{t+2}}^H} \left[ r(s_t, \mathbf{a}_t) + \gamma \cdot r(s_{t+1}, \mathbf{a}_{t+1}) + \gamma^2 \cdot r(s_{t+2}, \mathbf{a}_{t+2}) + \gamma^3 \cdot J_{s_{t+3}}^H(\pi_{s_{t+3}}^H) \right] \\
 &\vdots \\
 &\leq \mathbb{E}_{\mathbf{a}_t \sim \pi_{new}} \left[ \sum_{t=0}^{\infty} \gamma^t r(s_t, \mathbf{a}_t) \right] \\
 &= V^{\pi_{new}}(s_t).
 \end{aligned}$$

□

### A.3. Proof of Shared Optimum of Dual Policy Improvement

**Theorem 3.3** *Let  $\pi_0$  be an arbitrary initial policy. If the policy iteration using dual improvements is performed iteratively, the sequence of policies  $\{\pi_k\}_{k=0}^{\infty}$  converges to a unique optimal policy  $\pi_*$ . At the steady state, this optimal policy satisfies the following consistency relationship for all  $s_t \in \mathcal{S}$  and any arbitrary policy  $\pi$ :*

$$V^{\pi_*}(s) = J_s^H(\pi_*) = J_s^H(\pi_s^H) \geq J_s^H(\pi) \geq V^{\pi}(s). \quad (24)$$

*Proof. Part 1: Convergence.* Let  $\pi_i$  denote the policy at the  $i$ -th iteration. Since the value function sequence  $\{V^{\pi_i}(s)\}_{i=0}^{\infty}$  is monotonically non-decreasing and bounded above by  $R_{\max}/(1-\gamma)$ , it follows from the Monotone Convergence Theorem that the sequence converges to a limit  $V^{\pi_*}(s)$ , which corresponds to a steady-state policy  $\pi_*$ .

**Part 2: Fixed Point Consistency.** We next demonstrate that at the fixed point where  $\pi_{old} = \pi_*$ , the identity  $V^{\pi_*}(s) = J_s^H(\pi_s^H)$  holds. First, by definition,  $\pi_s^H$  is the optimal solution that maximizes  $J_s^H(\cdot)$ ; thus,  $V^{\pi_*}(s) \leq J_s^H(\pi_s^H)$  must hold. Conversely, since  $\pi_*$  is the fixed point of the dual improvement operator, we have  $\pi_{new} = \pi_{old} = \pi_*$ . The policy improvement property ensures  $V^{\pi_{new}}(s) \geq J_s^H(\pi_s^H)$ , which implies  $V^{\pi_*}(s) \geq J_s^H(\pi_s^H)$ . Combining these two inequalities yields the steady-state identity  $V^{\pi_*}(s) = J_s^H(\pi_s^H) = J_s^H(\pi_*)$ .

**Part 3: Global Optimality.** Let  $\pi$  be any arbitrary policy. From the optimality of  $\pi_s^H$  at the fixed point, we have  $V^{\pi_*}(s) = J_s^H(\pi_s^H) \geq J_s^H(\pi)$ . By recursively expanding  $J_s^H(\pi)$  through  $H$ -step Bellman backups:

$$\begin{aligned}
 V^{\pi_*}(s) &\geq J_s^H(\pi) = \mathbb{E}_\pi \left[ \sum_{k=0}^{H-1} \gamma^k r(s_{t+k}, a_{t+k}) + \gamma^H V^{\pi_*}(s_{t+H}) \right] \\
 &\geq \mathbb{E}_\pi \left[ \sum_{k=0}^{H-1} \gamma^k r(s_{t+k}, a_{t+k}) + \gamma^H \mathbb{E}_\pi \left[ \sum_{j=0}^{H-1} \gamma^j r(s_{t+H+j}, a_{t+H+j}) + \gamma^H V^{\pi_*}(s_{t+2H}) \right] \right] \\
 &= \mathbb{E}_\pi \left[ \sum_{k=0}^{2H-1} \gamma^k r(s_{t+k}, a_{t+k}) + \gamma^{2H} V^{\pi_*}(s_{t+2H}) \right] \\
 &\vdots \\
 &\geq \lim_{n \rightarrow \infty} \mathbb{E}_\pi \left[ \sum_{k=0}^{nH-1} \gamma^k r(s_{t+k}, a_{t+k}) + \gamma^{nH} V^{\pi_*}(s_{t+nH}) \right] \\
 &= \mathbb{E}_\pi \left[ \sum_{k=0}^{\infty} \gamma^k r(s_{t+k}, a_{t+k}) \right] = V^\pi(s).
 \end{aligned}$$

The last step follows from the fact that  $\gamma^{nH} V^{\pi_*}(s_{t+nH}) \rightarrow 0$  as  $n \rightarrow \infty$  for  $\gamma \in [0, 1)$ . This establishes the global optimality of  $\pi_*$ , completing the proof.  $\square$

## B. Practical Implementations of Harmonized Dual Policy Improvement

This appendix provides the detailed derivation of the HDPG solution presented in Theorem 3.4. We start from the primal minimax formulation in (13) and show how it reduces to a one-dimensional optimization problem that can be solved much efficiently.

### B.1. Equivalent Reformulation as a Linear Program

We first observe that the inner maximization loop in (13) can be equivalently rewritten as a linear program over a scalar variable  $\alpha \in [0, 2]$ :

$$\max_{\alpha \in [0, 2]} -\langle d(\alpha), h \rangle, \quad (25)$$

where

$$d(\alpha) = \alpha g_{\max Q} + (2 - \alpha) g_{\min KL}. \quad (26)$$

The equivalence follows because the maximum of a linear function over the convex hull of two points  $g_{\max Q}$  and  $g_{\min KL}$  is attained at one of the extremes, i.e., at  $\alpha = 0$  or  $\alpha = 2$  (Dantzig, 2016).

### B.2. Lagrangian Dual Transformation for the Trust Region Constraint

To obtain a more tractable convex form, we square the trust-region constraint (a standard technique in convex optimization (Boyd & Vandenberghe, 2004)), replacing  $\|h - \hat{g}\| \leq \rho \|\hat{g}\|$  with

$$\|h - \hat{g}\|^2 \leq \rho^2 \|\hat{g}\|^2. \quad (27)$$

Thus the primal problem becomes

$$\begin{aligned}
 \min_{h \in \mathbb{R}^n} \max_{\alpha \in [0, 2]} & -\langle d(\alpha), h \rangle \\
 \text{s.t. } & \|h - \hat{g}\|^2 \leq \rho^2 \|\hat{g}\|^2.
 \end{aligned} \quad (28)$$

Introducing a Lagrangian multiplier  $\varphi \geq 0$  for the inequality constraint yields the dual problem

$$\max_{\varphi \geq 0} \min_{h \in \mathbb{R}^n} \max_{\alpha \in [0, 2]} -\langle d(\alpha), h \rangle + \frac{\varphi}{2} (\|h - \hat{g}\|^2 - \rho^2 \|\hat{g}\|^2). \quad (29)$$

The following standard lemmas guarantee convexity and strong duality.

**Lemma B.1** (Convexity of the dual problem (Boyd & Vandenberghe, 2004)). *The dual problem (29) is convex, but its optimal value may not coincide with that of the primal problem unless strong duality holds.*

**Lemma B.2** (Slater condition and strong duality (Boyd & Vandenberghe, 2004)). *If a convex primal problem has a strictly feasible point (Slater’s condition), then strong duality holds.*

Lemma B.1 confirms that (29) such a dual problem is convex. To verify Slater’s condition for (28), note that the point  $h = (1 + \rho/2)\hat{g}$  is strictly feasible because

$$\|h - \hat{g}\|^2 = \|(\rho/2)\hat{g}\|^2 = (\rho^2/4)\|\hat{g}\|^2 < \rho^2\|\hat{g}\|^2.$$

Hence, by Lemma B.2, strong duality holds: the optimal value of (29) equals that of the primal problem (28), and the transformation is exact.

### B.3. Practical process for solving HDPG

We now derive the closed-form solution stated in Theorem 3.4. After the equivalent reformulations above, the problem (29) still involves three nested optimization loops. Because the objective is convex in  $h$  and concave in  $(\alpha, \varphi)$ , we can interchange the min and max operations.

**Step 1: Solve for  $h$ .** Interchanging the order gives

$$\max_{\alpha \in [0, 2]} \max_{\varphi \geq 0} \min_{h \in \mathbb{R}^n} -\langle d(\alpha), h \rangle + \frac{\varphi}{2} (\|h - \hat{g}\|^2 - \rho^2 \|\hat{g}\|^2). \quad (30)$$

For fixed  $\alpha$  and  $\varphi$ , the inner minimization over  $h$  is an unconstrained quadratic problem. Its optimum is attained at

$$h^*(\alpha, \varphi) = \hat{g} + \frac{d(\alpha)}{\varphi},$$

which we substitute back to eliminate the  $h$ -variable:

$$\max_{\alpha \in [0, 2]} \max_{\varphi \geq 0} -\langle d(\alpha), \hat{g} \rangle - \frac{1}{2\varphi} \|d(\alpha)\|^2 - \frac{\varphi\rho^2}{2} \|\hat{g}\|^2. \quad (31)$$

**Step 2: Solve for  $\varphi$ .** For fixed  $\alpha$ , the maximization over  $\varphi \geq 0$  in (31) is a concave one-dimensional problem. Setting the derivative to zero yields the optimal

$$\varphi^*(\alpha) = \frac{\|d(\alpha)\|}{\rho\|\hat{g}\|}.$$

Substituting  $\varphi^*(\alpha)$  back into (31) simplifies the expression to

$$\max_{\alpha \in [0, 2]} \langle d(\alpha), \hat{g} \rangle + \rho\|\hat{g}\|\|d(\alpha)\|. \quad (32)$$

**Step 3: Solve for  $\alpha$ .** The final step is the scalar optimization

$$\alpha^* = \arg \max_{\alpha \in [0, 2]} \langle d(\alpha), \hat{g} \rangle + \rho\|\hat{g}\|\|d(\alpha)\|,$$

which can be solved efficiently by a line search over  $[0, 2]$ .

**Step 4: Recover the harmonized gradient.** Using the relations  $h^* = \hat{g} + d(\alpha^*)/\varphi^*$  and  $\varphi^* = \|d(\alpha^*)\|/(\rho\|\hat{g}\|)$ , we obtain the harmonized gradient

$$h = \hat{g} + \frac{d(\alpha^*)}{\varphi^*} = \hat{g} + \rho \frac{\|\hat{g}\|}{\|d(\alpha^*)\|} d(\alpha^*). \quad (33)$$

This completes the proof of Theorem 3.4.

**Remark B.3** (Dimensionality reduction). The primal problem (13) is defined in the full parameter space of the policy, whose dimension  $n$  can be in the millions or billions. Solving such a high-dimensional constrained minimax problem at every iteration would be prohibitive. **In contrast, the simplified problem (32) requires optimizing only a single scalar  $\alpha$ , which drastically reduces the computational cost and makes harmonized gradient fusion practically feasible.**

## C. Experimental Details

### C.1. Benchmark Introduction

**DeepMind Control Suite.** We evaluate on the 7 most challenging tasks involving the `dog` and `humanoid` agents. These tasks fall into two categories: (1) *Standing tasks*, where the agent must maintain upright balance, and (2) *Moving tasks*, which additionally require the agent to move at a target velocity. For moving tasks, the reward is defined as the product of the standing reward and the forward velocity reward, i.e.,  $\text{Reward} = (\text{Standing reward}) \times (\text{Forward velocity reward})$ .



Figure 5. Dog



Figure 6. Humanoid

**Standing reward:** Encourages the agent to maintain an upright posture.

**Forward velocity reward:** Ensures the agent moves at the target speed (1 m/s for dog-walk, 3 m/s for dog-trot, 9 m/s for dog-run, 1 m/s for humanoid-walk and 10 m/s for humanoid-run).

**Humanoid Bench.** We consider 7 typical locomotion tasks involving a Unitree H1hand robot. This robot is initialized to a standing position, with random noise added to all joint positions during each episode reset.

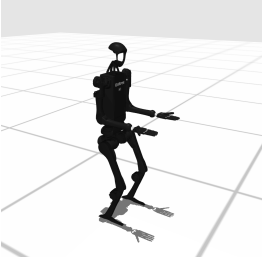


Figure 7. Stand

**Objective.** Maintain a standing pose.

**Reward:**  $R(s, a) = \text{stable} \times (0.5 \times \text{still}_x + 0.5 \times \text{still}_y)$ , where the `still` terms penalize non-zero velocities to encourage stationary balance. `stable` favors maintaining a stable and energy-efficient standing status.

**Termination.** 1000 steps, or when  $z_{\text{pelvis}} < 0.2$ .

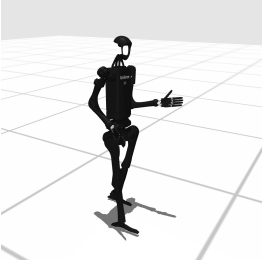


Figure 8. Walk

**Objective.** Keep forward velocity close to 1 m/s without falling to the ground.

**Reward:**  $R(s, a) = \text{stable} \times \text{tol}(v_x, (1, \infty), 1)$ , where `tol` encourages the agent to maintain a forward velocity  $v_x$  above 1 m/s, thereby promoting low-speed locomotion.

**Termination.** 1000 steps, or when  $z_{\text{pelvis}} < 0.2$ .

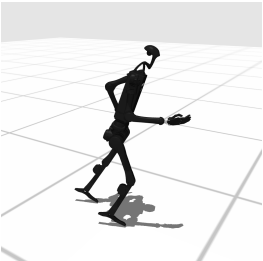


Figure 9. Run

**Objective.** Keep forward velocity close to 5 m/s without falling to the ground.

**Reward:**  $R(s, a) = \text{stable} \times \text{tol}(v_x, (5, \infty), 5)$ , where `tol` encourages the agent to maintain a forward velocity  $v_x$  above 5 m/s, thereby promoting high-speed locomotion.

**Termination.** 1000 steps, or when  $z_{\text{pelvis}} < 0.2$ .

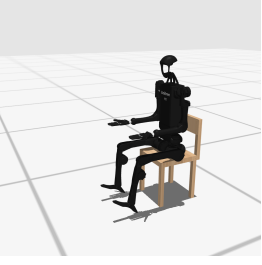


Figure 10. Sit

**Objective.** Sit onto a chair situated closely behind.

**Reward:**  $R(s, a) = (0.5 \cdot \text{sitting\_z} + 0.5 \cdot \text{sitting\_x} \cdot \text{sitting\_y}) \times \text{upright} \times \text{posture} \times e \times \text{mean}(\text{still\_x}, \text{still\_y})$ , where  $e$  is an energy penalty term,  $\text{sitting\_x}$ ,  $\text{sitting\_y}$ , and  $\text{sitting\_z}$  measure the robot’s positional tolerance relative to the chair.

**Termination.** 1000 steps, or when  $z_{\text{pelvis}} < 0.5$ .

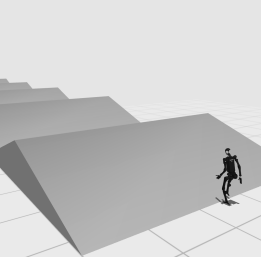


Figure 11. Slide

**Objective.** Walk over an iterating sequence of upward and downward slides at 1 m/s.

**Reward:**  $R(s, a) = e \times \text{tol}(v_x, (1, +\infty), 1) \times \text{upright} \times (\text{foot\_left} \times \text{foot\_right})$ , where  $\text{foot\_left}$  and  $\text{foot\_right}$  measure the vertical distance between the head and left/right foot respectively, ensuring proper foot positioning.

**Termination.** 1000 steps, or when  $z_{\text{proj}} < 0.6$ .

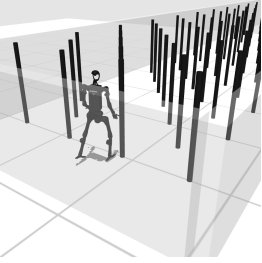


Figure 12. Pole

**Objective.** Travel forward over a dense forest of high thin poles, without colliding with them.

**Reward:**  $R(s, a) = \gamma_{\text{collision}} \times (0.5 \times \text{stable} + 0.5 \times \text{tol}(v_x, (1, +\infty), 1))$ , where the collision penalty  $\gamma_{\text{collision}}$  equals 0.1 if the robot collides with a pole, and 1 otherwise.

**Termination.** 1000 steps, or when  $z_{\text{pelvis}} < 0.6$ .



Figure 13. Hurdle

**Objective.** Keep forward velocity close to 5 m/s without falling to the ground.

**Reward:**  $R(s, a) = \text{stable} \times \text{tol}(v_x, (5, \infty), 5) \times \gamma_{\text{collision}}$ , which penalizes colliding with hurdle.

**Termination.** 1000 steps.

## C.2. Additional Comparison with Diffusion RL Baselines

**QVPO** (Ding et al., 2024): A generative policy baseline that utilizes  $Q$ -weighted variational loss to train diffusion-based policies for multi-modal exploration. **QSM** (Psenka et al., 2024): A method that utilizes  $Q$ -Score Matching to align the policy’s score function with the gradient of the learned  $Q$ -function. **DACER** (Wang et al., 2024): A framework that leverages  $Q$ -gradients to update the diffusion policy in an end-to-end manner. **DPMD** (Ma et al., 2025): An efficient diffusion RL algorithm that reweights the denoising score matching loss to perform mirror descent updates, achieving high-expressivity policy refinement without expensive backpropagation through the diffusion chain.

Table 2. Performance comparison on 7 DMC Suite tasks and 7 Humanoid Benchmark (H-Bench) tasks. Mean  $\pm$  Std over 5 seeds. **Bold** = best, underlined = second-best. Higher is better. (1) **H**- prefix refers to the *Humanoid* domain (e.g., H-Stand is Humanoid-Stand). (2) **D**- prefix refers to the *Dog* domain (e.g., D-Trot is Dog-Trot). (3) **H1**- prefix denotes the *H1hand* humanoid benchmark.

| DMC Suite | H-Stand         | H-Walk           | H-Run            | D-Stand           | D-Walk           | D-Trot           | D-Run            | Average          |
|-----------|-----------------|------------------|------------------|-------------------|------------------|------------------|------------------|------------------|
| QVPO      | 5.3 $\pm$ 0.0   | 1.8 $\pm$ 0.1    | 1.2 $\pm$ 0.0    | 111.1 $\pm$ 13.2  | 19.9 $\pm$ 0.7   | 16.5 $\pm$ 8.4   | 43.2 $\pm$ 4.9   | 27.6 $\pm$ 2.8   |
| QSM       | 8.6 $\pm$ 0.3   | 2.2 $\pm$ 0.1    | 1.5 $\pm$ 0.0    | 405.4 $\pm$ 10.8  | 106.4 $\pm$ 13.8 | 78.9 $\pm$ 9.0   | 64.4 $\pm$ 4.8   | 95.3 $\pm$ 5.6   |
| DPMD      | 9.1 $\pm$ 0.1   | 2.9 $\pm$ 0.7    | 1.5 $\pm$ 0.0    | 382.2 $\pm$ 48.3  | 47.6 $\pm$ 6.5   | 51.4 $\pm$ 5.7   | 29.0 $\pm$ 17.1  | 70.3 $\pm$ 15.7  |
| DACER     | 8.2 $\pm$ 0.6   | 2.6 $\pm$ 0.1    | 1.5 $\pm$ 0.1    | 409.5 $\pm$ 114.8 | 150.4 $\pm$ 60.4 | 125.1 $\pm$ 42.3 | 131.1 $\pm$ 8.8  | 116.9 $\pm$ 30.1 |
| BOOM-H    | 948.6 $\pm$ 2.2 | 942.0 $\pm$ 18.5 | 496.1 $\pm$ 47.0 | 987.6 $\pm$ 3.8   | 971.6 $\pm$ 13.4 | 915.0 $\pm$ 22.2 | 707.5 $\pm$ 10.7 | 852.6 $\pm$ 16.8 |

| H-Bench | H1-Stand          | H1-Walk         | H1-Run           | H1-Sit            | H1-Slide         | H1-Pole          | H1-Hurdle        | Average          |
|---------|-------------------|-----------------|------------------|-------------------|------------------|------------------|------------------|------------------|
| QVPO    | 11.4 $\pm$ 0.2    | 5.6 $\pm$ 0.1   | 3.9 $\pm$ 0.7    | 8.7 $\pm$ 1.6     | 4.5 $\pm$ 0.1    | 20.1 $\pm$ 2.5   | 5.5 $\pm$ 0.0    | 6.8 $\pm$ 0.8    |
| QSM     | 31.9 $\pm$ 2.5    | 12.1 $\pm$ 2.0  | 6.6 $\pm$ 0.6    | 55.5 $\pm$ 11.4   | 22.7 $\pm$ 2.8   | 63.0 $\pm$ 11.3  | 9.5 $\pm$ 1.2    | 28.6 $\pm$ 4.6   |
| DPMD    | 75.3 $\pm$ 7.4    | 51.2 $\pm$ 4.1  | 15.8 $\pm$ 1.8   | 321.2 $\pm$ 139.4 | 44.5 $\pm$ 4.7   | 93.6 $\pm$ 34.3  | 15.6 $\pm$ 6.1   | 81.0 $\pm$ 5.3   |
| DACER   | 311.1 $\pm$ 276.2 | 44.6 $\pm$ 1.0  | 21.8 $\pm$ 4.7   | 309.4 $\pm$ 186.6 | 58.5 $\pm$ 4.2   | 112.2 $\pm$ 32.5 | 24.1 $\pm$ 11.0  | 124.3 $\pm$ 69.3 |
| BOOM-H  | 942.8 $\pm$ 11.9  | 929.4 $\pm$ 1.6 | 752.1 $\pm$ 27.9 | 911.6 $\pm$ 7.3   | 610.5 $\pm$ 33.2 | 823.2 $\pm$ 39.0 | 270.8 $\pm$ 65.0 | 746.1 $\pm$ 30.0 |

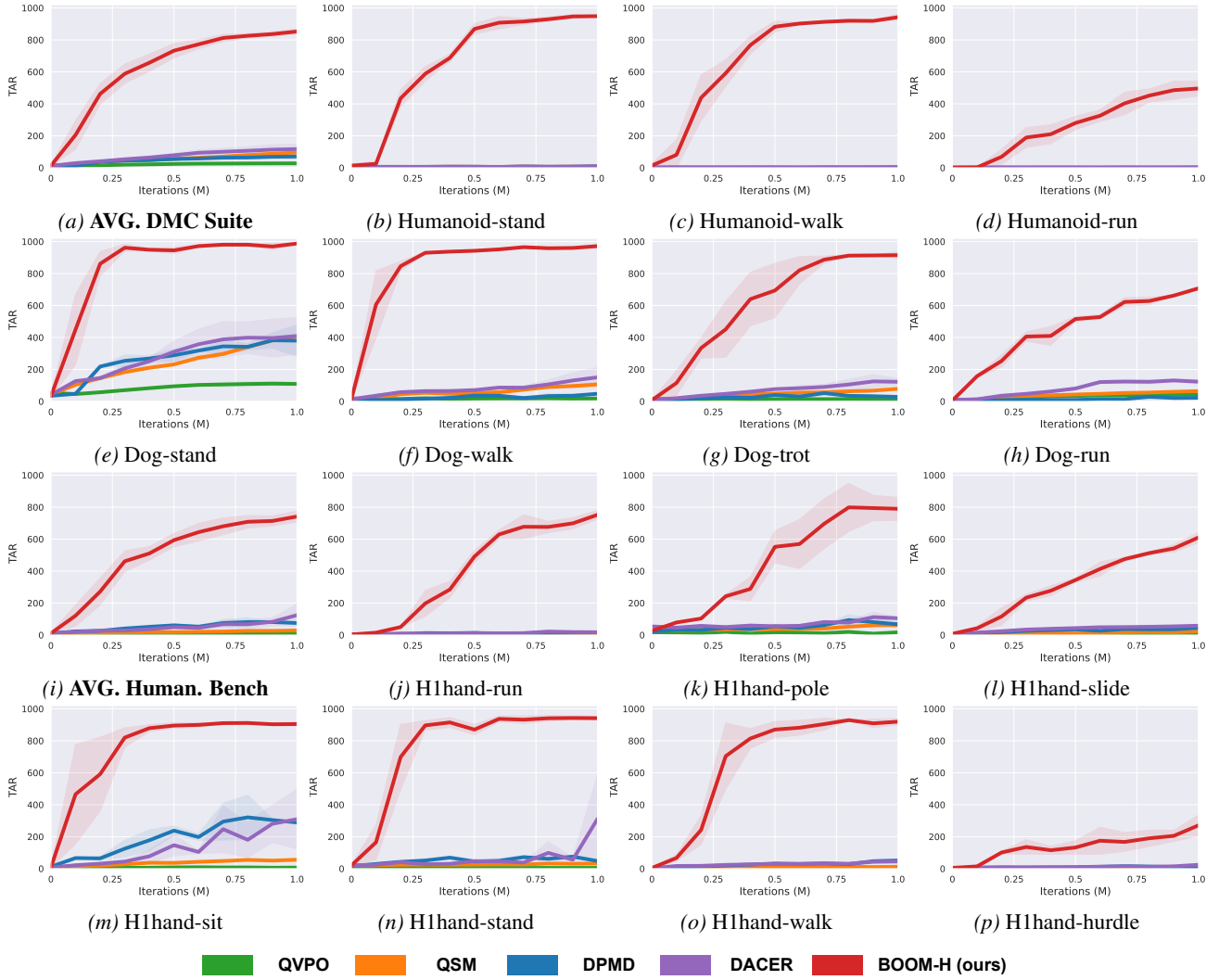


Figure 14. **Training curves on benchmarks.** The solid lines represent the mean, while the shaded regions indicate the confidence interval over three runs. The average performance curves appear at the left corner of the 1st and 3rd rows, respectively, highlighted in **bold**.

### C.3. Reproducibility Statement & Detailed Hyperparameters

The details of hyperparameter settings are listed in Table 3. In this paper, we evaluate our algorithm for each tasks over five random seeds. The CPU used is the AMD Ryzen Threadripper 3960X 24-Core Processor, and the GPU used is NVIDIA GeForce RTX 3090Ti. For all baselines, we align the fundamental settings including iteration steps and evaluation protocol for a fair comparison.

For SAC, DreamerV3 and TDMPC2, we report the baseline results released by TDMPC2, which are obtained from the official repositories of [https://github.com/denisyarats/pytorch\\_sac](https://github.com/denisyarats/pytorch_sac), <https://github.com/danijar/dreamerv3>, and <https://github.com/nicklashansen/tdmpc2> respectively. For BMPC, we use the official implementation <https://github.com/wertyuilife2/bmpc>. For DSACT, we use the official implementation <https://github.com/Jingliang-Duan/DSAC-v2>. For QVPO, we use the official implementation <https://github.com/wadx2019/qvpo>. For QSM, we use the official implementation [https://github.com/escontra/score\\_matching\\_rl](https://github.com/escontra/score_matching_rl). For DPMD, we use the official implementation [https://github.com/mahaitongdae/diffusion\\_policy\\_online\\_rl](https://github.com/mahaitongdae/diffusion_policy_online_rl). For DACER, we use the official implementation <https://github.com/happy-yan/DACER-Diffusion-with-Online-RL>. For BOOM, we use the official implementation [https://github.com/molunitu/BOOM\\_MBRL](https://github.com/molunitu/BOOM_MBRL).

Table 3. Hyperparameter settings.

| Hyperparameter                                      | Value                            | Hyperparameter                      | Value              |
|-----------------------------------------------------|----------------------------------|-------------------------------------|--------------------|
| <b>Training</b>                                     |                                  |                                     |                    |
| Learning rate                                       | $3 \times 10^{-4}$               | Target network update rate          | 0.5                |
| Encoder learning rate                               | $1 \times 10^{-4}$               | Discount factor ( $\gamma$ )        | 0.99               |
| Sample batch size                                   | 1                                | Gradient Clipping Norm              | 20                 |
| Replay batch size                                   | 256                              | Optimizer                           | Adam               |
| Buffer size                                         | 1,000,000                        | Scale norm                          | Moving (5%, 95%)   |
| Steps                                               | 2,000,000                        | Sampling                            | Uniform            |
| <b>World Model</b>                                  |                                  |                                     |                    |
| Reward loss coefficient ( $c_r$ )                   | 0.1                              | Dynamics loss coefficient ( $c_f$ ) | 20                 |
| Value loss coefficient ( $c_q$ )                    | 0.1                              | Value functions esemble             | 5                  |
| Number of value bins                                | 101                              | Warmup steps                        | 5000               |
| <b>Planner</b>                                      |                                  |                                     |                    |
| MPPI Iterations                                     | 6 (8 if $\ \mathcal{A}\  > 20$ ) | Minimum planner std                 | 0.05               |
| Population size                                     | 512                              | Maximum planner std                 | 2                  |
| Number of elites                                    | 64                               | Horizon                             | 3                  |
| Policy prior samples                                | 24                               |                                     |                    |
| <b>Actor</b>                                        |                                  |                                     |                    |
| Minimum policy log std                              | -10                              | Entropy coefficient ( $\alpha$ )    | $1 \times 10^{-4}$ |
| Maximum policy log std                              | 2                                | Harmonization coefficient $\rho$    | 0.01               |
| Alignment coefficient $\lambda$                     | $\ \mathcal{A}\ /1000$           |                                     |                    |
| <b>Architecture (around 5M parameters in total)</b> |                                  |                                     |                    |
| Encoder layers                                      | 2                                | Latent space dimension              | 512                |
| Encoder dimension                                   | 256                              | Task embedding dimension            | 96                 |
| MLP hidden layer dimension                          | 512                              | Q function drop out rate            | 0.01               |
| MLP activation                                      | Mish                             | MLP Normalization                   | LayerNorm          |